

Parameter	Fixed Bias (Base bias)	Emitter Feedback Bias	Collector Feedback Bias	Voltage Divider Biasing
Circuit Diagram				
Step 1A.	-	-	-	To obtain Thevenin Equivelaent Circuits and Draw Self Bias Circuit
Step 1 :- Apply KVL to Base Loop & Find IBQ	$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B}$	$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta_{dc})R_E}$	$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta_{dc})R_C}$	$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta_{dc})R_E}$
Step 2 :- Find $I_{CQ} = \beta_{dc} I_{BQ}$	$I_{CQ} = \beta_{dc} I_{BQ}$	$I_{CQ} = \beta_{dc} I_{BQ}$	$I_{CQ} = \beta_{dc} I_{BQ}$	$I_{CQ} = \beta_{dc} I_{BQ}$
Step 3 :- Apply KVL to Collector loop & Find VCEQ	$V_{CEQ} = V_{CC} - I_{CQ}R_C$	$V_{CEQ} = V_{CC} - I_{CQ}R_C - I_ER_E$	$V_{CEQ} = V_{CC} - I_{CQ}R_C - I_BR_C$	$V_{CEQ} = V_{CC} - I_{CQ}R_C - I_ER_E$
Step 4 :- Q Point (VCEQ , ICQ)	$Q(V_{CEQ}, I_{CQ})$	$Q(V_{CEQ}, I_{CQ})$	$Q(V_{CEQ}, I_{CQ})$	$Q(V_{CEQ}, I_{CQ})$
Step 5 :- Stabilisation of Q-point	Poor Stability	Stability better than Fixed Bias Circuit	Stability better than Fixed Bias Circuit & Emitter Feedback Bias circuit	Very Good Stability
Stability Factor	$s = 1 + \beta_{dc}$	$s = \frac{1 + \beta_{dc}}{1 + \beta_{dc} \frac{R_E}{R_E + R_B}}$	$s = \frac{1 + \beta_{dc}}{1 + \beta_{dc} \frac{R_C}{R_E + R_C}}$	$s = \frac{1 + \beta_{dc}}{1 + \beta_{dc} \frac{R_E}{R_E + R_B}}$
Feedback Mechanism	No Feedback	Feedback due to RE Resistor	Feedback due to RC Resistor	Feedback due to RE Resistor
Applications	Used in application where stability is not important	Used in Switching application	Used in Switching application	Used in aplication where stability is most important